Maker Guide

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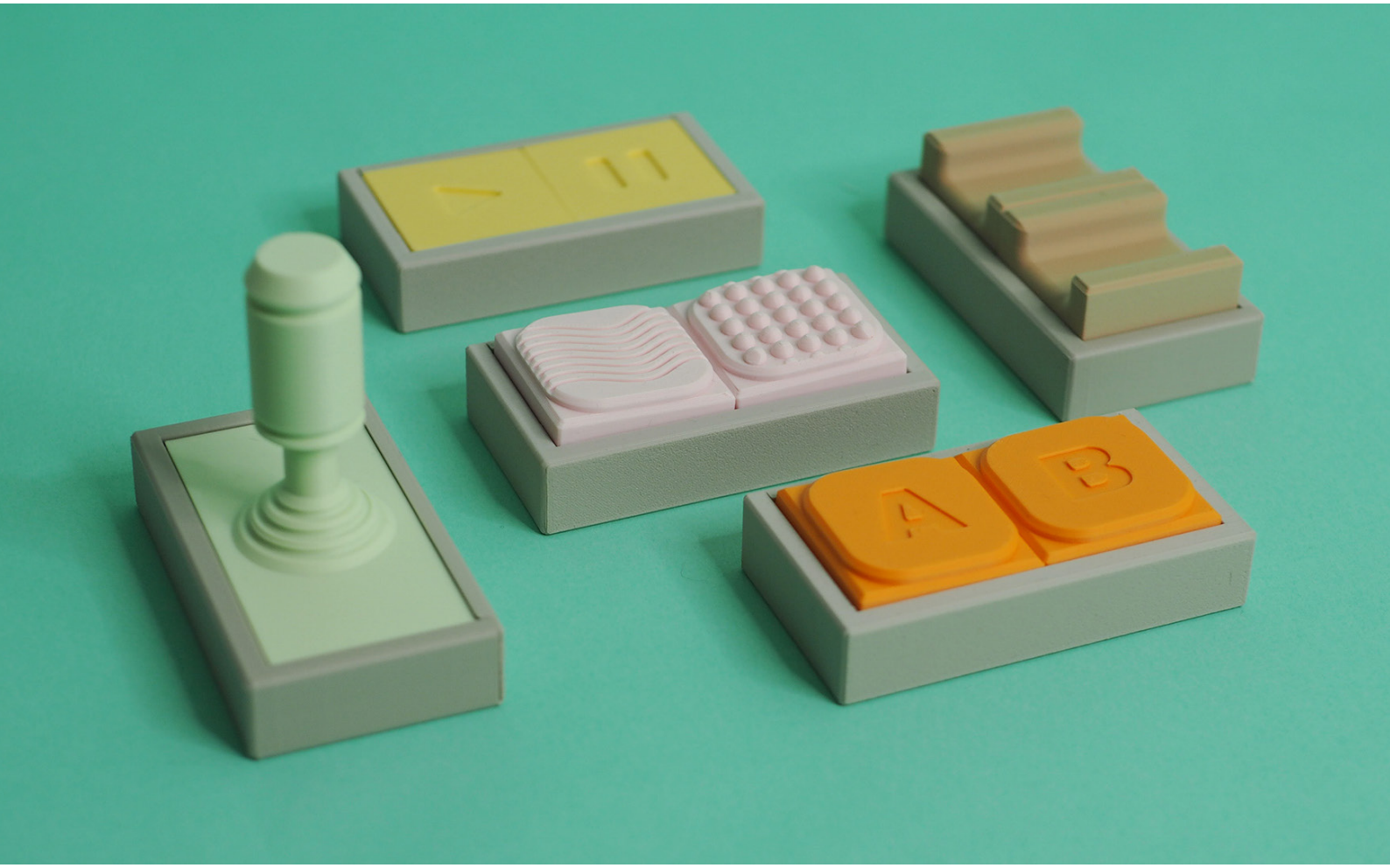


Tap is a bluetooth dual switch access device that enables people with disabilities to control mobile phones, tablets and computers. It works by activating switch control on the selected device, and mapping Tap's customisable buttons to the desired functions - whether this be 'move to next item' and 'tap', or media functions such as 'play' and 'pause'.

why?

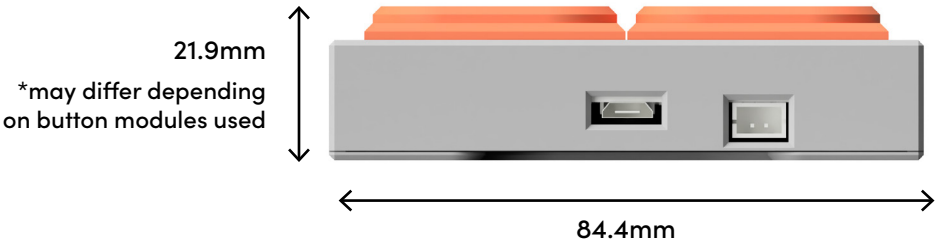
Computer devices and systems are vital for communication, education, employment, social interaction, and more. But more often than not, they can be inaccessible for people with disabilities.

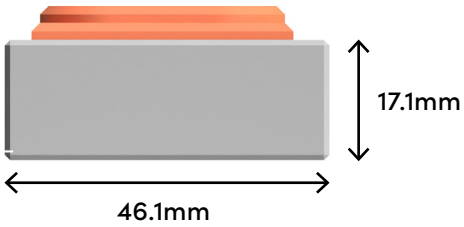
Tap is designed for people who have limited hand mobility and struggle to use traditional input methods for computer devices, such as keyboard and mouse, small buttons or touchscreens.

It aims to empower individuals with physical and cognitive impairments to interact with technology, engage with their surroundings, and enhance their independence.

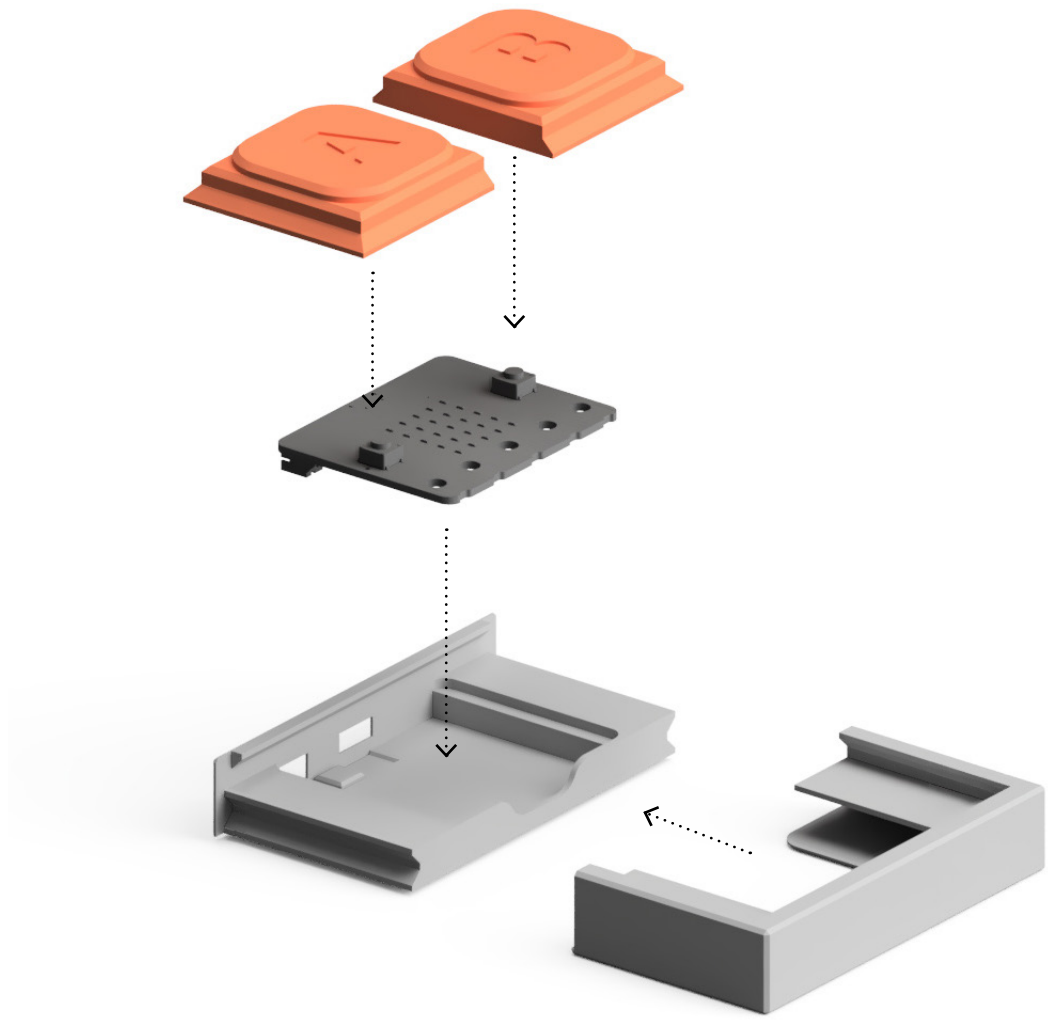
Unlike other existing solutions in the market, Tap is **affordable**, **highly customisable** and **open source**.

A
B


21.9mm
*may differ depending
on button modules used
84.4mm


17.1mm
46.1mm

≈Weight: 45g



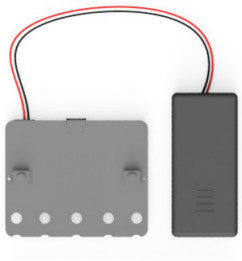
how?

A micro:bit v2 pocket-sized computer sits at the heart of the device. It is programmed so that when the bluetooth connection is active and the A button is pressed, it sends the keyboard 'tab' key as an output. Similarly, when the B button is pressed, it sends the keyboard 'enter' key as an output.

By activating switch control on any android, iOS or chromebook device, these inputs can be mapped to various functions such as 'move to next item' and 'tap', or media functions such as 'play' and 'pause'.

The micro:bit's existing buttons are transformed into larger accessible buttons through the use of 4 3D printed components. These include a case, sliding lid, and 2 buttons that sit directly on top of the existing buttons. When pushed, they simply press down and activate the existing buttons.

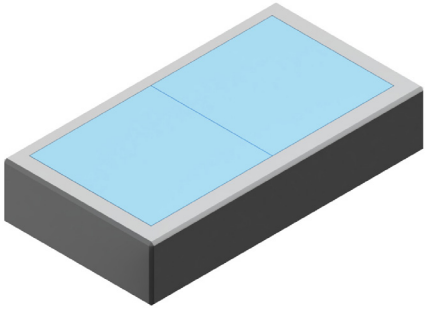
materials

Description	Qty Required	≈ Cost (USD)	
Micro:bit v2 with Battery Pack *Alternatively, you can replace the battery pack with a USB type-B cable to power from your device.	1	$19.95	 Example Option: https://bit.ly/4djc88l
3D Printed Cradle	1	$0.30 (PLA material cost)	
3D Printed Sliding Lid	1	$0.33 (PLA material cost)	
3D Printed A Button *Multiple options available - see next page	1	$0.17 (PLA material cost)	
3D Printed A Button *Multiple options available - see next page	1	$0.17 (PLA material cost)	

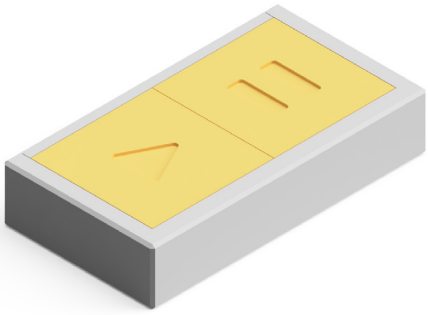
button options

The following button options are available for you to choose from in the model files folder.

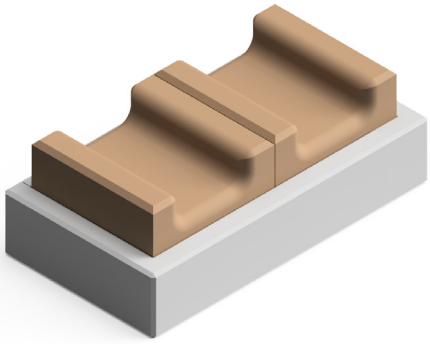
A/B Buttons (Raised)



Plain (Flat)

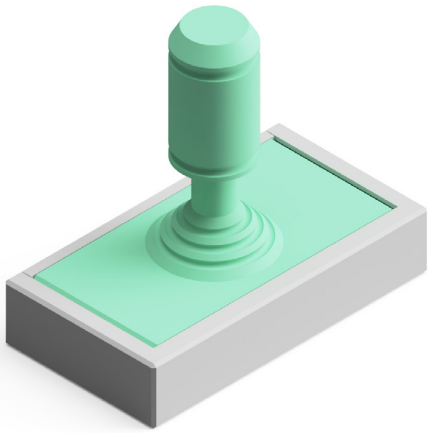


Play/Pause (Flat)



Finger Grooves (Raised)

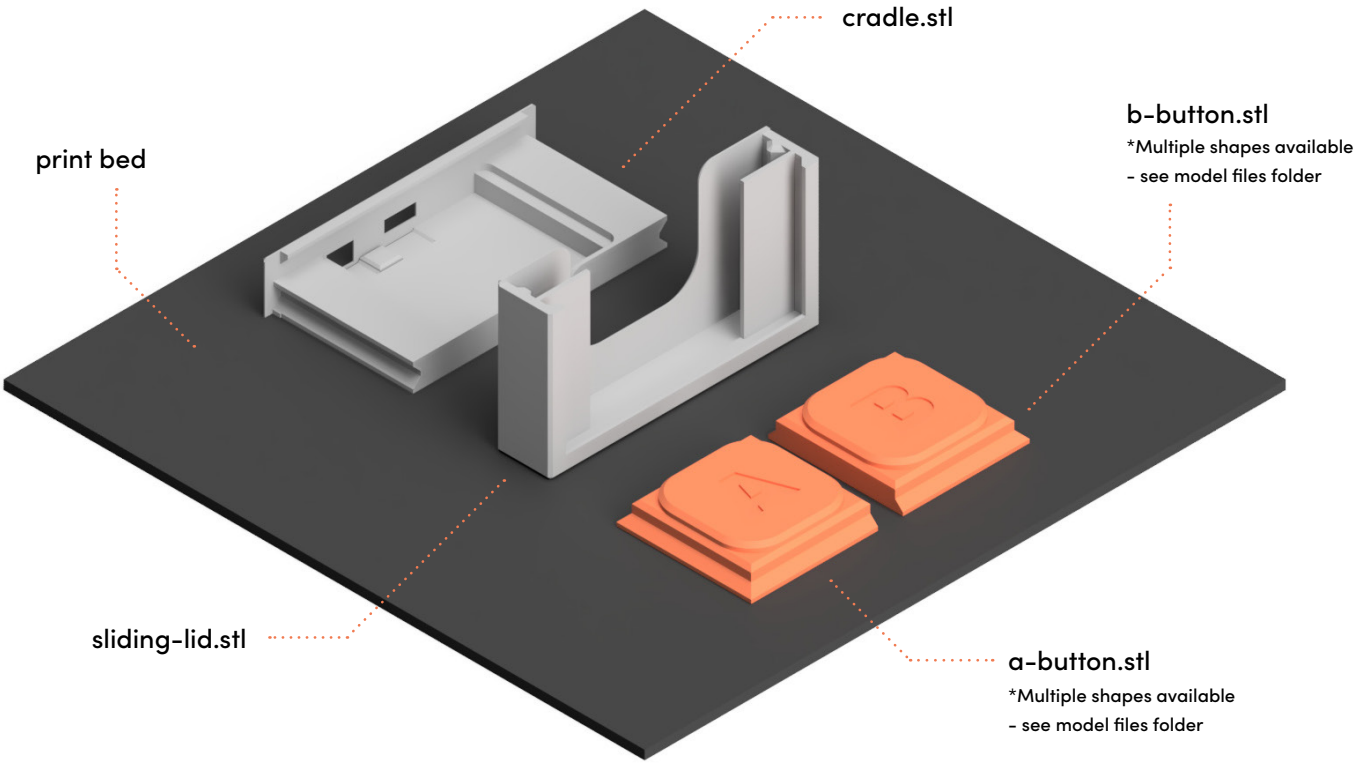
Textures (Raised)



Joystick *Coming soon

3d printing

Please follow the below guidance on orientation and print settings.



Settings

Technology	FFF
Material	PLA
Nozzle Diameter	0.4mm
Layer Height	0.1-0.15mm
Infill	15%
Support	None

The device will still be functional with layer heights of 0.2mm+. However, we recommend 0.1-0.15 for an improved tactile experience and guaranteed activation across the entire surface area of buttons.

Estimated Stats

Print Time	3 hours
Mass	37g
Material Cost	$0.97 (USD)

disclaimer

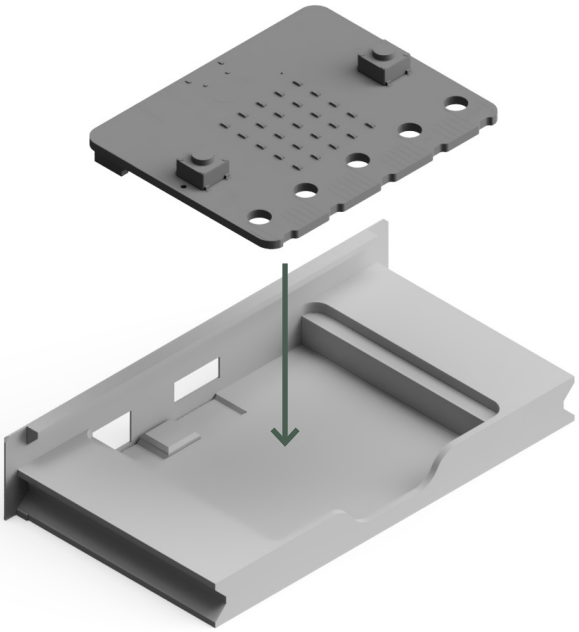
Due to the varying tolerances of 3D printers and material properties, we cannot guarantee the same functionality or tactile experience for each manufactured Tap device.

If you have any issues with 3D printing or usage, please contact hello@weareprintlab.com and we will do our best to advise you with alternative print settings or design adjustments.

assembly

01.

Connect your micro:bit to a computer via USB connection. Then copy the tabEnterOutput.hex file from the model files folder and paste it onto the micro:bit, just like you would copy and paste a file onto a USB stick!

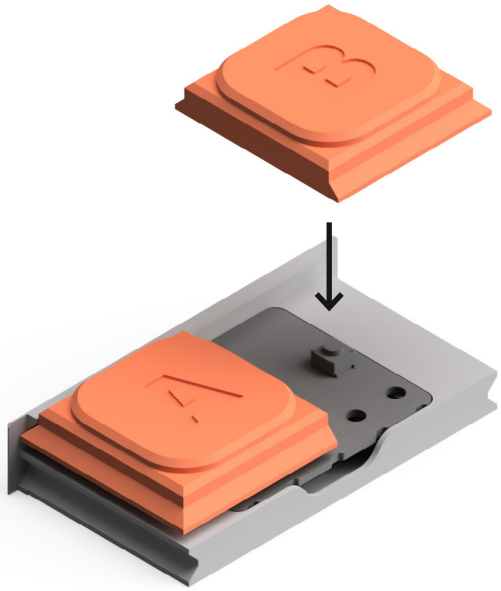


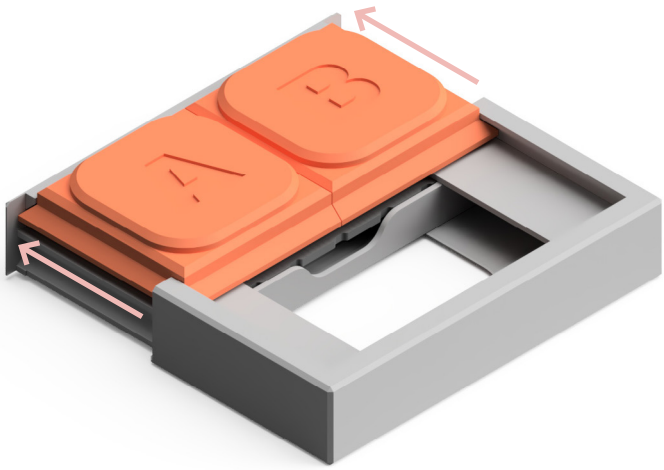
02.

Place the micro:bit into the 3D printed cradle with the LED side pointing upwards. When placing, tilt the micro:bit so the connection ports feed through the holes in the cradle first.

03.

Place the 3D printed A and B buttons on top of the micro:bit's existing switches. Ensure you line them up so the hole on the underside of the 3D printed buttons sit over the micro:bit switches.



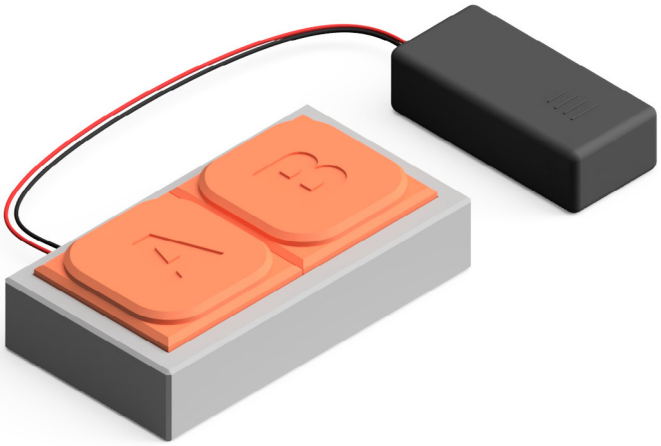


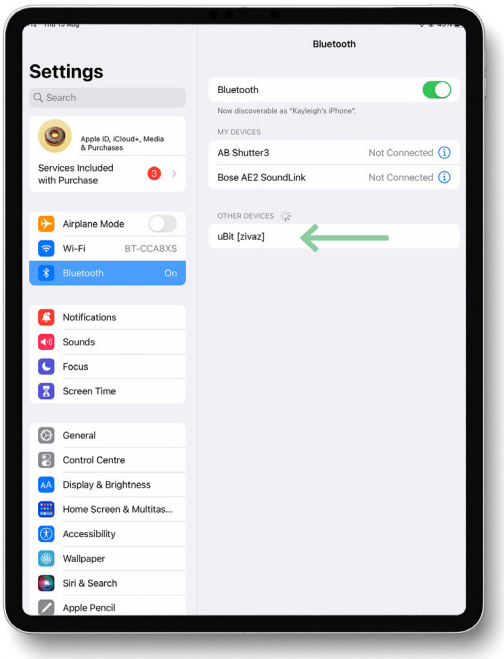
04.

Push the 3D printed sliding lid onto the device. Refer to the diagram opposite to guide you, ensuring the lid 'traps' the buttons from falling out.

05.

Time to power up! Plug in the micro:bit's battery pack, or alternatively, use a USB type-B cable to power it from your device.





06.

Go to your device and pair the micro:bit in your bluetooth settings. The micro:bit will be named uBit.

switch access

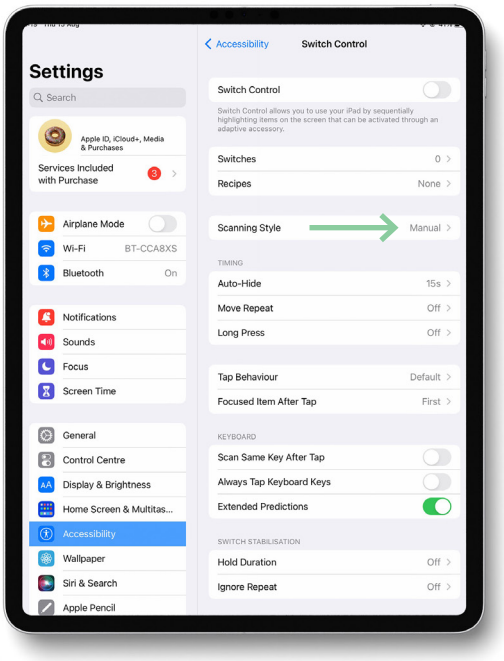
The next part of the assembly instructions involves activating switch access on your mobile, tablet or computer device. The instructions in this guide will show you how to set up 'manual scanning' on an iPad device with iOS. This will map the A button to the action 'Move to Next Item', and the B button to 'Tap'. If you are using a different device, the steps will be similar, and below are some external links with further information.

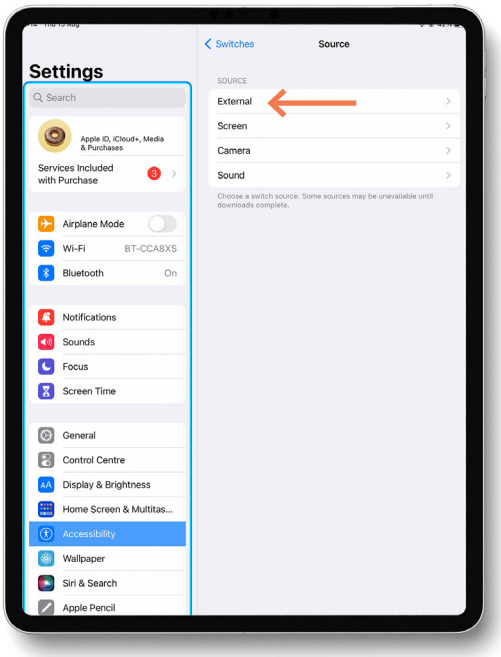
Switch Access for Android: https://bit.ly/3X4ZxjQ
Switch Access for Chromebook: https://bit.ly/4dr5AVG

Depending on your requirements, you can also map the A and B buttons to different actions, such as controlling media with play and pause actions. Feel free to experiment with the options on your device!

07.

On your iOS device, go to Settings > Accessibility > Switch Control and ensure scanning style is set to Manual.



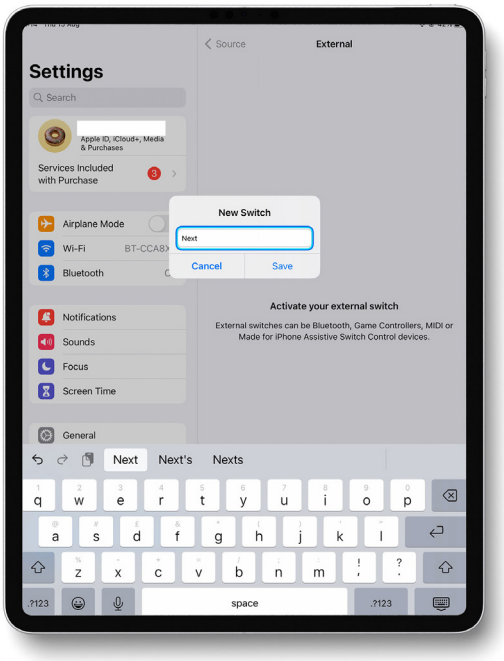


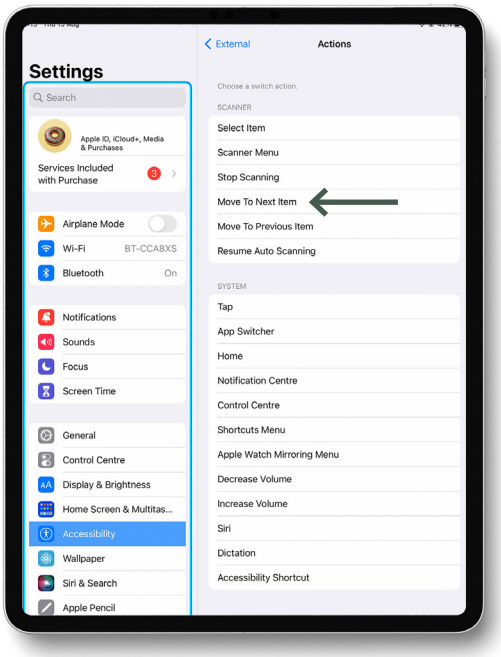
08.

In the Switch Control menu you are already in, go to Switches > Add New Switch > External.

09.

You will now be prompted to activate your external switch. Simply press the A button on your Tap device and name it as ‘Next’ and press Save.



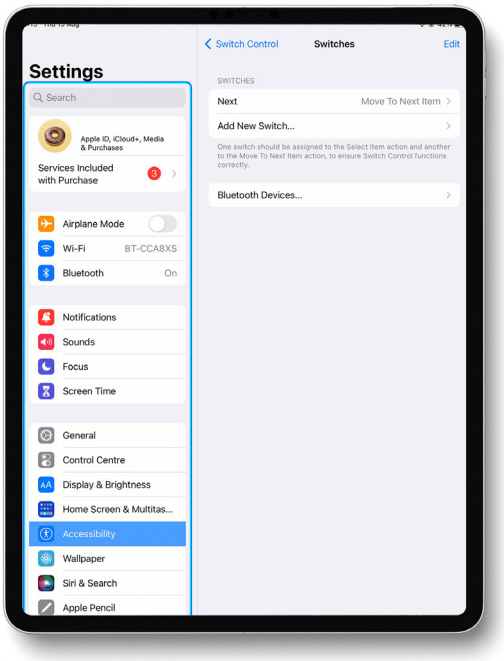


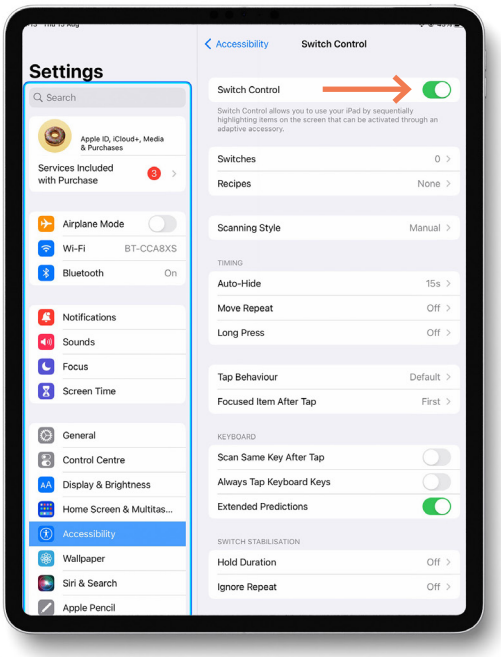
10.

You can now select which action you’d like your A button to activate. Select ‘**Move to Next Item**’.

11.

Button A is now mapped! Repeat steps 8-10 for the B button but name the switch as ‘Tap’ and map it to the ‘tap’ action.





12.

The final step is to ensure Switch Control is toggled on and then you’re all set!

usage

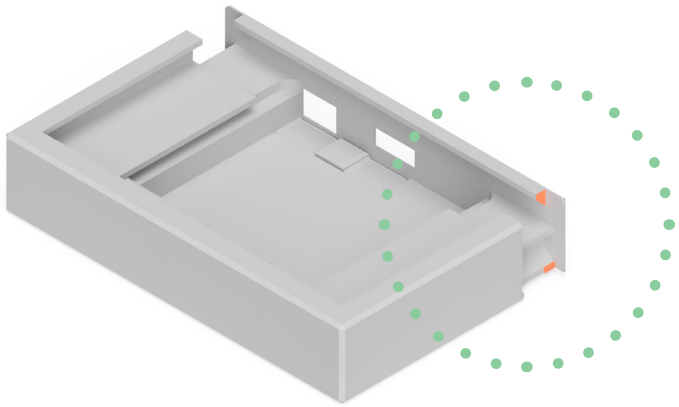
When switch control is activated, you'll see a blue border highlighting an item or a group of items. Pressing the A button will change what items are highlighted and once you reach the desired items, press the B button to select. To power off Tap, press and hold the flexure button on the underside of the device for 5 seconds. To power it up again, press the same button once.

adjustments

Due to the varying tolerances of 3D printers, you may find that the sliding lid is either too loose or too tight. If this is the case, follow the below instructions to edit the 3D CAD model and reprint the cradle.

How the Lid Works

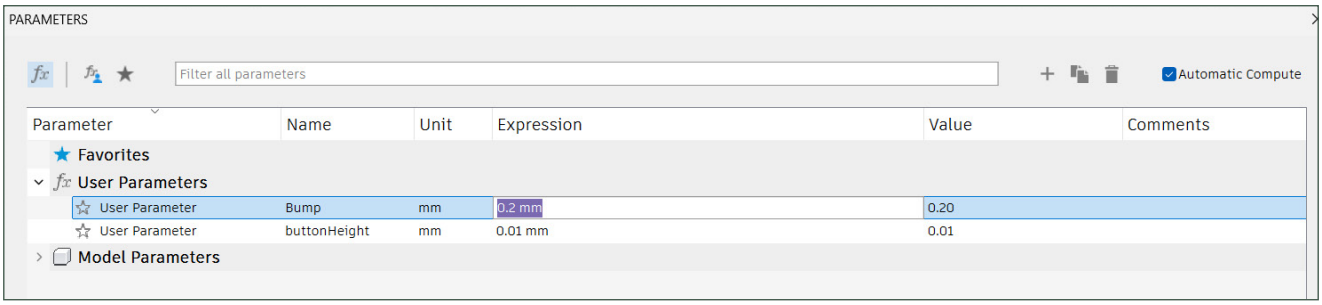
The lid was designed with a relatively loose fit when it slides over the buttons. As it reaches the top of the cradle, it meets 2 parts on each side (highlighted below), which stick out and apply tight friction to keep the lid in place during use. We call these parts 'bumps', and the distance they stick out is easily editable.



Editing the Bump Value

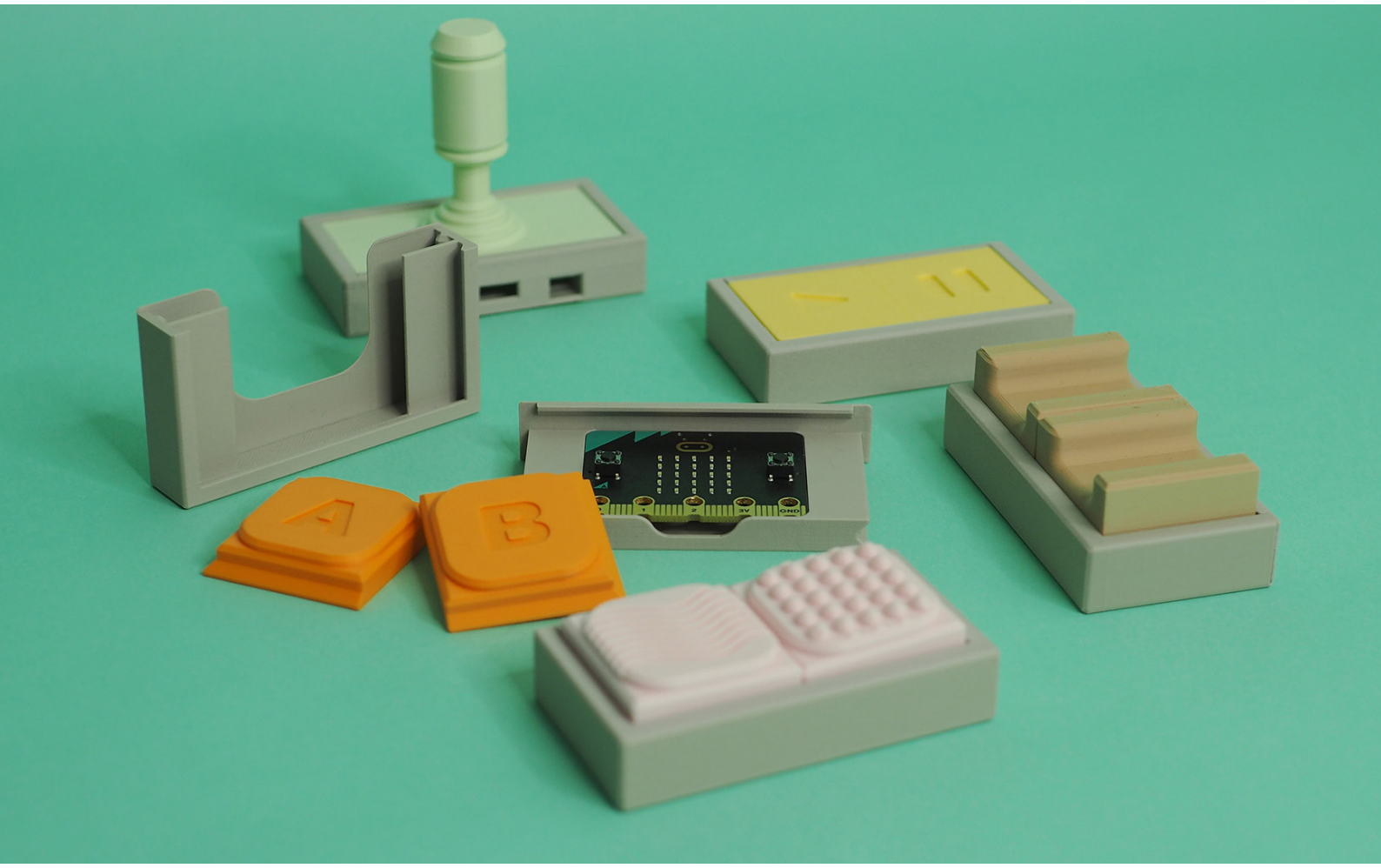
The default bump value is 0.4mm. If the sliding lid was a little loose, we recommend changing this to 0.6mm. And if it was a little tight, we recommend changing it to 0.2mm.

To change the bump value, download Autodesk Fusion (free for education and personal use) and open the file Tap.f3d from the model files folder. In the modify menu, select Change Parameters. Then click in the Bump Expression field and change the value. Changing the value will automatically update the design with either a larger or smaller bump.



Once the bump value is changed, you can right-click on the Cradle body in the browser on the left side of the screen and select 'Save as Mesh', before saving out a new STL file to 3D print and test.

evolution



The Tap design files have been released under a Creative Commons CC BY SA 4.0 License, whereby we encourage people to modify, adapt and improve the design as they see fit. We are particularly interested in physical modifications, such as mounting solutions, as well as code modifications that might integrate additional functionality. If you have any feedback or suggestions, please email them to hello@weareprintlab.com.

learn

Want to learn how to use 3D CAD and 3D printing to design your own unique version of Tap? Check out the full course on PrintLab, where we guide you through each step of the design process to make a custom version for a specific end user of your choice.

Course Link: www.weareprintlab.com/projects/at-dual-switch

www.weareprintlab.com